

Mesna

Mesna is a drug given with some types of chemotherapy to protect the bladder from irritation.

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Why mesna is given

The chemotherapy drugs [ifosfamide](#) and [cyclophosphamide](#) can irritate the lining of the bladder. This can cause bleeding from the bladder, which may lead to blood in your urine (pee). This is called haematuria. Mesna helps to protect your bladder and prevent irritation and bleeding.

People being treated with ifosfamide will always have mesna.

Cyclophosphamide only causes bleeding from the bladder when it is given in high doses. Most people who have treatment with cyclophosphamide do not have a high dose. This means they will not need mesna. Your doctor, nurse or pharmacist can tell you if you are having a high dose of cyclophosphamide.

While you are having mesna, your urine is tested for signs of blood. If you have blood in your urine, you will be given extra mesna.

Doing these things can help protect your bladder during treatment:

- Drink plenty of water – at least 2 litres (3½ pints) daily. Your nurse can tell you how much you should drink.
- Empty your bladder often during the treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](https://www.medicines.org.uk/emc) website, which has patient information leaflets (PIL) for individual drugs.

How mesna is given

Mesna can be given in the following ways:

- As a drip (infusion) – you have this into a vein at the same time as your chemotherapy.
- As a liquid that you drink – this is added to a flavoured soft drink, such as orange juice or cola. Your doctor, nurse or pharmacist can explain more about having mesna as a drink.
- As a tablet – it is important to take mesna tablets exactly as your doctor, nurse or pharmacist explains.

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Your course of treatment

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Side effects while treatment is being given

Some people may have side effects while they are being given the chemotherapy or shortly after they have it:

Allergic reactions

Rarely, some people have an allergic reaction while having this treatment. Allergic reactions are most common with the first few doses, but can happen at any time during treatment. A reaction is usually mild.

Signs of a reaction include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Pain or irritation to the vein

If you have mesna as a drip (infusion), it can cause a reaction at the place where it is given or further along the vein. Symptoms of a reaction include:

- pain
-

a rash

- red or darker skin
- swelling
- itchy skin where the drip is given.

If you have these symptoms, tell your nurse straight away. They can check the site.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Feeling sick

You may feel sick while having this treatment. But your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. If you [feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible.

If you are sick immediately after taking mesna tablets, contact the hospital for advice. It is important you do not miss a dose of mesna .

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
-

any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Skin changes

This treatment can affect your skin. Skin changes can include:

- a rash or bumps
- dry or itchy skin
- areas of red or darker skin
- blisters.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Flu-like symptoms

This treatment may cause flu-like symptoms such as:

- feeling hot, cold or shivery
- a high temperature or fever
- a headache
- aching.

If you have flu-like symptoms, you should contact the hospital straight away on the 24-hour number.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

If you have muscle or joint pain, try:

- placing a heat pad or covered hot water bottle against the painful area
- taking warm baths
- planning your activities to include regular rests.

Behaviour changes

Mesna can cause difficulty sleeping and nightmares. It may also make it hard for you to concentrate. If you have any of these side effects, tell your doctor, nurse or pharmacist. They may make some changes to your treatment if the side effects become a problem.

Effects on the nervous system

Rarely, this treatment can affect the brain and nerves. Contact the hospital if you have any of these symptoms during treatment or after it ends:

- any weakness, numbness or tingling in your arms, legs or face
- a headache that does not get better
- dizziness
- drowsiness or confusion
- problems with moving or speaking
- any unexplained pain in other parts of your body
- fainting or loss of consciousness.

Eyesight changes

This treatment can affect your eyes and eyesight. Contact the hospital if you have any of these symptoms during treatment or after it ends:

- sore, painful, or swollen eyes
- eyes that are sensitive to light
- changes to your eyesight.

You may need treatment to help these symptoms. A specialist eye doctor (ophthalmologist) may also need to check your eyes.

Effects on the heart

This treatment can affect how the heart works.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat • swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Sore mouth

This treatment may cause a [sore mouth](#). You may also have bleeding gums. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth affects how much you can drink or eat
- your mouth, tongue or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness
- pain when breathing.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Other information

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

Pregnancy and breastfeeding

Ask your doctor or pharmacist for advice before taking this medicine if you:

- are pregnant
- think you may be pregnant
- are planning to have baby.

You are also advised not to [breastfeed](#) when taking this treatment. This is because the medicine may be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly

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make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help